Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

In Situ TAC

Objective Analysis products

INSITU_GLO_PHY_TS_OA_NRT_013_002

Issue: 1.5

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CHANGE RECORD

Issue	Date	§	Description of Change	Author	Checked By
1.0	18/01/2017	All	Creation of the document	T. Szekely	L. Petit de la Villéon
1.1	13/03/2020	All	New Version of the ISAS software and of the climatology first guess	D. Dobler	L. Petit de la Villéon
1.2	25/11/2022	All	Dataset name change Document template change	T. Szekely	S. Tarot
1.3	28/08/2023	All	Dataset Update	T. Szekely	S. Tarot
1.4	19/06/2024	All	Extension to 5500 m depth	T. Szekely	S. Tarot
1.5	27/10/2025	All	Update of the First guess climatology	T. Szekely	S. Bengoufa

ACRONYM TABLE

Acronym	Description
Argo	International profiling float network (www.argo.net)
CORA	Coriolis Ocean Dataset for Reanalysis
CTD	Conductivity, Temperature, and Depth sensor
FSD	Falst Salty Drift
HST	Halosteric height anomaly
ISAS	In Situ Analysis System
LOPS	Laboratoire d'Océanographie Physique et Spatiale
MBT	Mechanical bathythermographs
OA	Objective analysis
PCTVAR	percentage of the product parameter variance associated to the in-situ measurements
PSU	PSU
QC	Quality Checks
RMS	Root Mean Square
SSS	Sea Surface Salinity
XBT	expandable bathythermographs

KEY DEFINITIONS

NA

RELEVANT DOCUMENT LIST

NA

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

The document describes the quality of the global gridded temperature and salinity fields based on the near real time temperature and salinity product INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030.

The details of the product INSITU_GLO_PHY_TS_OA_MY_013_002 are given in Table 1.

***Table 1**: Short description of the product INSITU_GLO_PHY_TS_OA_NRT_013_002.*

Product Specification name	INSITU_GLO_PHY_TS_OA_MY_013_002
Short description	Objective analysis of monthly export from Coriolis database
Variables	Temperature (TEMP), salinity gridded field (PSAL), percentage of the variance (PCTVAR) and parameter error
Geographical coverage	Global (-77°N to 89.9°N, -180°E to 179.5°E)
Spatial resolution	0.5° in longitude and varying in latitude with 0.5° at equator and 0.1° at 77°S / 0.2° at 89.9°N (see Figure 1)
Vertical levels	There are 187 vertical levels, starting at 0 meters. From 3 meters up to 100 meters in depth: every 5 meters; from 110 meters to 800 meters in depth: every 10 meters; from 820 meters to 2000 meters in depth: every 20 meters; from 2100 to 5500 meter in depth: every 100 meters
Temporal resolution	Monthly fields, centred on the 15th day of each month Covering the period January 2023 to present
Target delivery time	Monthly
Update frequency	When substantial changes/improvements have taken place
Data source	Temperature and salinity profiles from INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030

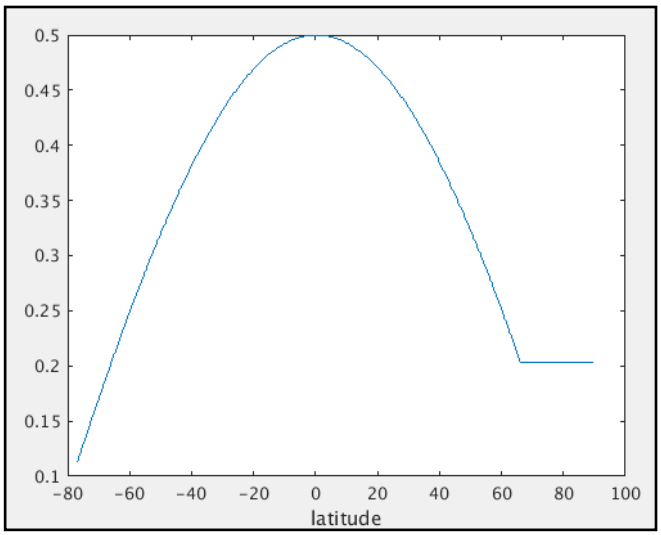

***Figure 1**: Resolution of the product (in degrees) as a function of latitude.*

I.2 Summary of the results

This product is based on the objective analysis (i.e., the interpolation of sparse in-situ observations on a regular grid) of the temperature and salinity measurements taken from the global near real time dataset from the Coriolis Database (INSITU_GLO_PHYBGCWAV_DISCRETE_MYNRT_013_030 product) and linearly interpolated on 187 vertical levels between 0 and 5500 meters. The objective analysis is performed using the In Situ Analysis System (ISAS) tool, based on the work by Gaillard et al. (2015) and developed by N. Kolodziejczyk and A. Prigent[1]. This tool is based on the method developed by Bretherton et al. (1976). The method interpolates the temperature and salinity profiles in three dimensional fields, taking into account the correlation between nearby profiles, the bathymetry and the variability of the Rossby radius.

The accuracy of the method depends on the spatial resolution of the initial temperature and salinity sampling. Consequently, the results of the gridded fields are very close to the first guess in the poorly sampled zone.

To avoid misinterpretations of the gridded temperature and salinity fields, the product includes parameter error fields that consider the coverage. The error fields are a composite of estimated ocean variability in the sparsely sampled zones and measurement errors in the well sampled zones.

The estimated accuracy numbers (EANs) of this product show a mean PCTVAR almost constant during all the time period covered by the product. It thus shows the consistency of the product (see section I.3 for more details).

The change of configuration (section IV.1) has been shown to have a limited influence on the dataset results, and the product has been shown to be consistent with the associated MY product and with reference products for halosteric sea level anomaly (section II.2) and global ocean heat content (section IV.2).

[1] https://www.seanoe.org/data/00412/52367/

I.3 Estimated Accuracy Numbers

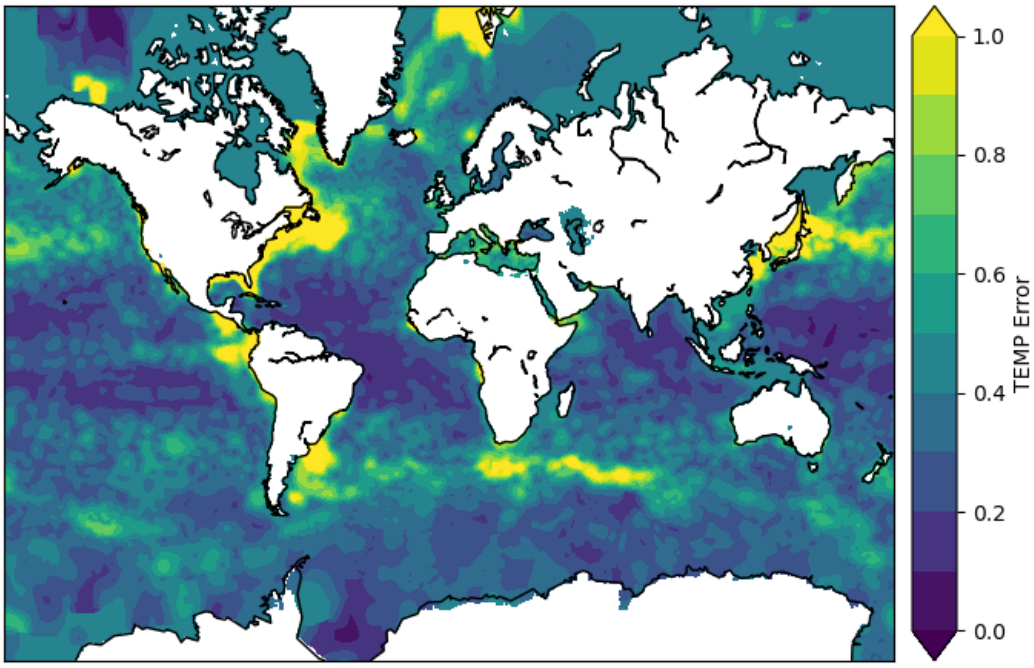

Figure 2: *Ocean surface temperature error (units: °C) in January 2023 for the INSITU_GLO_PHY_TS_OA_NRT_013_002 product.*

Figure 2 gives an overview of the surface temperature error field for the NRT product in January 2023. The parameter error depends both on the local variability and the availability of the data. For instance, the low parameter error south of the Kerguelen islands (49°S, 69°E) is associated with sea mammal borne measurements. See Figure 3 for a zoom-in on this area.

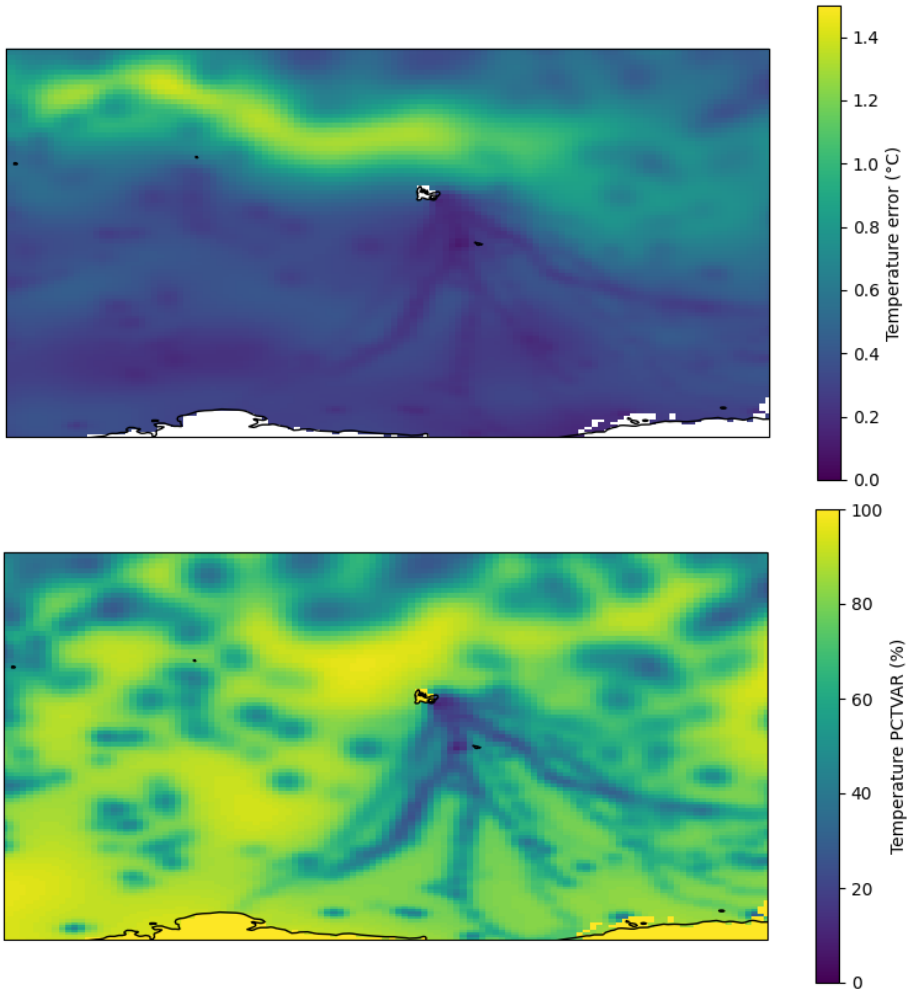

Figure 3: *Temperature error (top, units: degrees Celsius). Temperature field PCTVAR (bottom, unitless). For an area south of the Kerguelen islands (49°S, 69°E)Fields are dated January 2023.*

The percentage of the product parameter variance associated with the in-situ measurements parameter (PCTVAR) is distributed along with the analysed field (TEMP or PSAL) and the error parameter. The value of PCTVAR is 100 when there are no measurements within 1 correlation length of the grid point, and close to 0 when there are plenty of in-situ measurements in the vicinity of the grid point.

Figure 4 shows the time evolution for different depths of the percentage of surface field with a PCTVAR lower than 80 % for temperature and salinity fields. The sea water pump of the ARGO floats´ CTD sensors are cut off in the top five meters of the ocean to avoid damage to the conductivity sensor. As a consequence, the data coverage is lower at zero-meter depth than at five-meter depth.

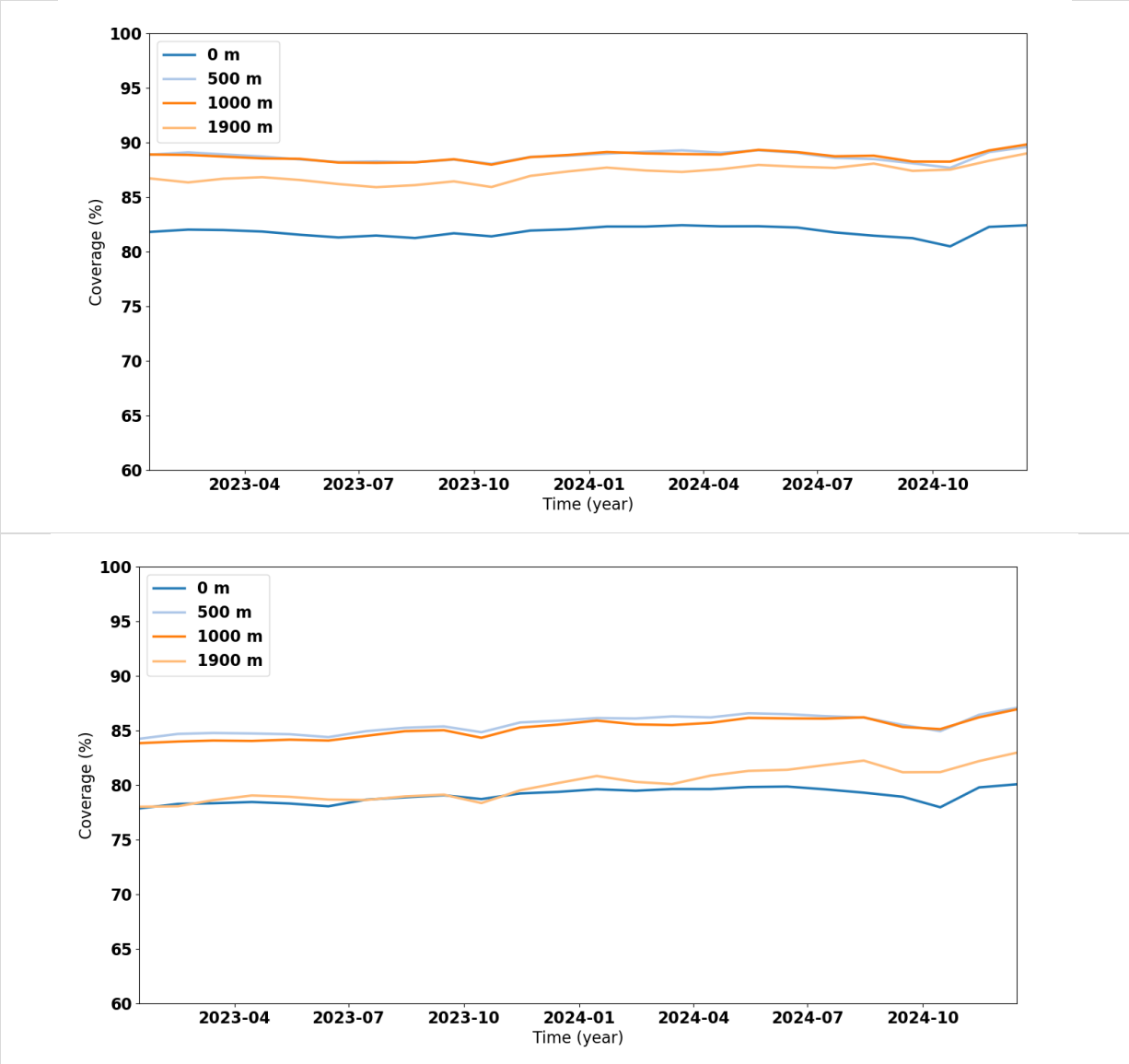

Figure 4: *Percentage of surface field with PCTVAR (percent variance field, which express the proportion of total variance explained by the observations) < 80% for NRT temperature (top) and salinity (bottom) fields for different depth levels.*

II PRODUCTION SYSTEM DESCRIPTION

Production centres name: Ifremer

Poduction system name: Global ocean In-Situ temperature and salinity, delayed time mode validated, objective analysis (INSITU_GLO_PHY_TS_OA_NRT_013_002)

Production System description:

The INSITU_GLO_PHY_TS_OA_NRT_013_002 product provides an objective analysis of temperature and salinity profiles distributed within it. This analysis was generated using version 8 of the ISAS tool (Gaillard et al., 2015). The temperature and salinity measurements are interpolated onto a regular grid using the optimal interpolation technique introduced by Bretherton et al. (1976). To conform to this approach, the ISAS tool initially calculates temperature and salinity anomalies relative to climatology. These anomalies are then mapped onto the regular grid, and the result is combined with the reference climatology to create the final dataset.

In the equations developed by Bretherton et al. (1976), interpolation errors tend to be lower when the mean of the anomalies is minimized. Therefore, it is crucial to utilize climatologies that closely align with the observed data. Given the context of increasing ocean heat content, using a set of climatologies is essential to minimize interpolation errors. The choice of climatology (first-guess) for the objective analysis varies depending on the year of analysis. The specific list of climatologies and their corresponding analysis years is provided in the product PUM document.

The interpolated profiles are mostly ARGO floats (Wong et al, 2020), and sea mammal measurements (Treasure et al, 2017), conductivity temperature depth profiles (CTD), XBTs and mechanical bathythermographs measurements.

II.1 ISAS Calculation method

This product is composed of temperature and salinity measurements collected from different sources by the Coriolis data center[2]. Most of the data concerned by the objective analysis are ARGO profiles, XBTs, CTDs, MBTs and sea mammals borne measurements. However, only the profiles having both temperature and salinity are included in order to work with consistent fields of temperature and salinity. Given the better quality checks (QC) and coverage with Argo, XCTD are now discarded, as the number of XCTD in the years 2018-2023 was remarkably low compared to the ARGO, Sea mammals or CTD profiles. The effect of the XCTD rejection on the product overall quality is very light since there is only a few dozen XCTD profiles deployed in the recent years compared to the roughly 12 000 monthly ARGO profiles.

First, the temperature and the salinity profiles are linearly interpolated to fit the ISAS grid vertical sampling (see Table 1 for the vertical sampling scheme description). The extrapolated points with a low confidence level are flagged after QC. The second set of QC warnings is triggered for profiles that deviate from the first guess by more than six standard deviations of the temperature or salinity first guess field. This step aims to exclude profiles with significant data-climatology anomalies in low-variability zones while retaining them in high-variability zones.

[2] https://www.coriolis.eu.org

The objective analysis parameters are summarized in Equation (1), named correlation function:

$$C(dx, dy, dt) = \sum_{i=1}^{2} \sigma_{Li}^2 \exp\left(\frac{dx^2}{2L_{ix}^2} + \frac{dy^2}{2L_{iy}^2} + \frac{dt^2}{2L_{it}^2}\right) \quad (1)$$

With *dx*, *dy*, *dt* being the distance between a profile and a grid point, $\sigma_{L1} = \sigma_{ocean}$ and $\sigma_{L2} = 2\ \sigma_{ocean}$.

The correlation scales are L_{1x} = 300 km, L_{2x} = Rossby radius (Feliks et al. 1985) , L_{1y} = 300 km, L_{2y} = Rossby radius and $T_{1,2}$ = 21 days.

The objective analysis parameters are summarized in Table 1 (see Gaillard et al., 2015 for details). The objective analysis is performed on measurement anomalies relative to a reference field, the first guess, at the 15th day of each month. The first guesses are based on the ISAS 2017 monthly climatology and the ISAS version 7.0 is used.

Data with high spatial and/or temporal resolution are often subject to averaging in order to prevent an artificial bias toward specific measurements. This operation becomes necessary because the equation developed by Bretherton et al. (1976), assumes that all analyzed observations are independent. Consequently, when both high-frequency and low-frequency profilers fall within the correlation scales, as described in Equation (1), they may carry similar weights in Equation (1). As a result, the solution may be unduly influenced by the high-resolution profiles rather than the sparser ones. To mitigate this issue, mooring data is averaged using a 9-day criterion, while other datasets, such as sea mammal-borne profiles and ice-tethered profiles, are averaged using a 5-day/5 km criterion. The criterion thresholds have been chosen to optimize the product data sampling and the product stability.

Finally, the temperature and salinity grids are reconstructed by summing the objective analysis of the anomalies and the first guess field, following the method of Gaillard et al. (2015).

This method produces monthly gridded fields of temperature and salinity and the associated PCTVAR fields. The PCTVAR fields are a gridded quantity varying between 0 and 100, and are related to the influence of the observations on the interpolated field. A PCTVAR value that tends toward zero can be considered uncorrelated with the first guess. On the contrary, a PCTVAR of 100 is associated to a final value equal to the first guess (such as at high latitudes where few observations are available, (see Figure 5). Consequently, the user should keep in mind that the accuracy of the global objective analysis solution is better in deeper ocean zones and north of 60°S. Note also that small differences can occur between temperature and salinity PCTVAR, since drifting Argo salinity profiles can be discarded by quality process.

Figure 5 shows an example of an analyzed field map at surface for salinity (left) and temperature (right) and the corresponding PCTVAR values.

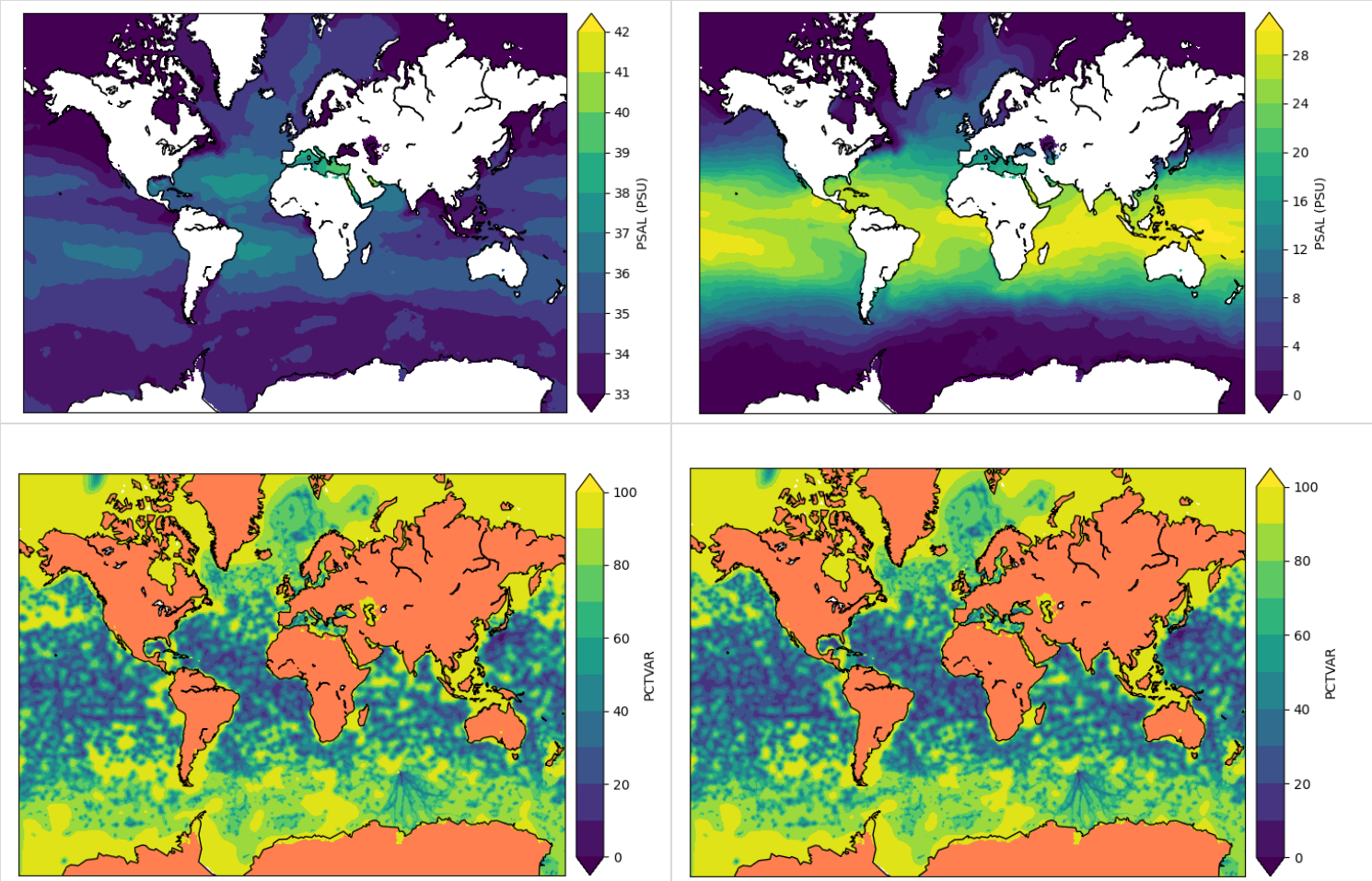

***Figure 5**: Surface salinity analysed field (top left, units: PSU) and associated PCTVAR (bottom left, unitless) and surface temperature analysed field (top right, units: °C) and associated PCTVAR (bottom right, unitless), in January 2023.*

II.2 Correction of the ARGO fast salinity drift

Since 2016, many of the ARGO profiles are affected by a drift of the salinity sensor (hereafter Fast Salty Drift, or FSD). According to the ARGO global data centre, this problem may appear on up to 25% of the ARGO floats, leading to a positive drift of the salinity in ARGO based gridded products after 2016.

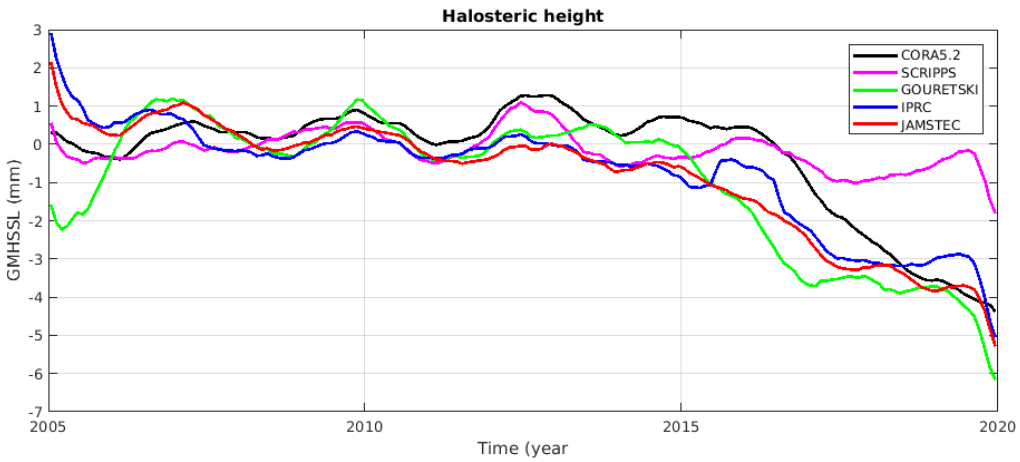

***Figure 6**: Halosteric height anomaly (HAS) for CORA 5.2 (December 2020 version) and gridded products from SCRIPPS[3], IPRC[4], JAMSTEC[5] and Goureski[6].*

Figure 6 shows that all objective analysis (OA) tested products except for those of SCRIPPS have a Halosteric height anomaly (HAS) drift beginning in 2015-2016, from -4 mm (CORA 5.2, December 2020 version). Copernicus marine product: INSITU_GLO_PHY_TS_DISCRETE_MY_013_001) to -6 mm (Gouretski[5]) drift in 2020. The SCRIPPS[3] product differs from the others since it is the only product where ARGO real time profiles are adjusted to a climatology before the objective analysis.

The CORA product appears to have a limited drift compared to other products. In fact, the validation process of CORA data flags a fraction of the drifting floats. In addition, the CORA OA production framework is synchronized to the ARGO delayed time mode validation and adjustment process, in order to integrate a maximum of the ARGO delayed time mode adjusted profiles.

Furthermore, an adjustment method (described in Roemmich and Gilson, 2009) is employed to correct the ARGO profiles, addressing drifting profiles that may go unnoticed within the ARGO framework. This algorithm aligns with the one utilized in the SCRIPPS product.

For each ARGO real time profile, the algorithm calculates the anomaly between the measurements and a climatology based on the OA products for 2005-2015. If the mean anomaly is lower than 0.25 PSU (practical Salinity Unit), the profile is adjusted using the value of the anomaly, otherwise the profile is flagged as “bad” in the objective analysis and excluded from the calculation.

[3] https://sio-argo.ucsd.edu/RG_Climatology.html
[4] http://apdrc.soest.hawaii.edu/datadoc/argo_iprc_gridded.php
[5] https://www.jamstec.go.jp/argo_research/dataset/moaagpv/moaa_en.html
[6] Gouretski refers to EN4 objective analysis with Gouretski and Cheng, 2020 XBT correction scheme.
Data access : https://hadleyserver.metoffice.gov.uk/en4/download.html

Figure 7 shows that the adjustment method eliminates almost all of the drift. The correction statistics are then analysed to flag the remaining FSD profiles in CORA. The adjustment is not applied before 2016 because the differences between the OA and OA adjusted curves before 2016 are caused by the anomaly period calculation and the ARGO global DAC listing removal. The list of rejected profiles with the higher adjustment values is quality controlled, and flagged when necessary.

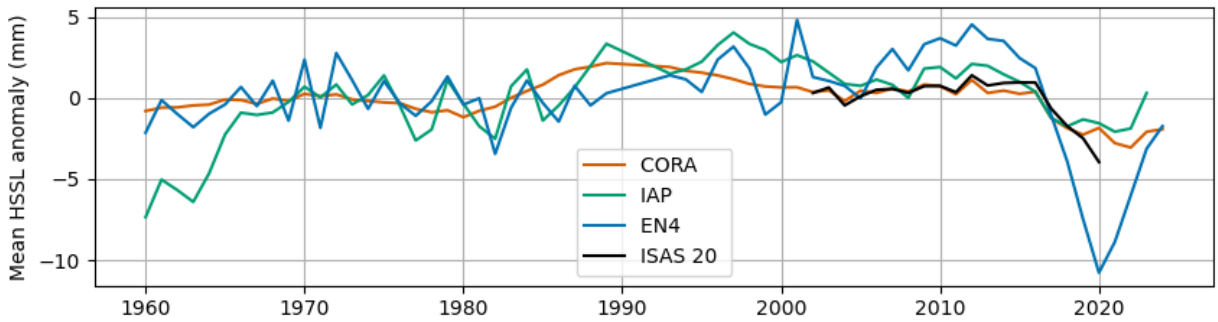

***Figure 7**: Halosteric height anomaly (HSA) for CORA 5.2 (November 2025 version) and gridded products from IAP[7], EN4[8] and ISAS 20[9].*

[7] https://essd.copernicus.org/articles/16/3517/2024/essd-16-3517-2024-discussion.html
[8] https://www.metoffice.gov.uk/hadobs/en4/
[9] https://www.seanoe.org/data/00412/52367/

III VALIDATION FRAMEWORK

The dataset has been validated by comparison with the previously used ISAS release and configuration (release 6.2 and first guess ISAS11). The main differences between the 6.2 and 7.0 releases are:

- Correction in the computation of super-observations
- Treatment of platform as a whole
- Tuning of alert system configuration
- Optimal Analysis configuration variance parameters

The following diagnostics have been performed:

- Comparison of salinity fields and temperature fields
- Comparison of global heat content results
- Comparison of global salt content results

These diagnostics have been performed using several test cases:

- Old 6.2 release/ISAS11 first-guess
- 7.0 release/ISAS13 first-guess
- LOPS ISAS15 Argo only product (Kolodziejczyk et al., 2017)
- CORA 5.2 product (Szekely et al., 2019).

In the following section, some validation results are presented. A comparison of monthly fields of the former and the latest version of the objective analysis (section IV.1), and a validation of the global ocean heat content in the objective analysis fields (section IV.2). A presentation of the global mean halosteric anomaly of the product has already been presented in section II.2.

IV VALIDATION RESULTS

IV.1 Comparison of salinity fields and temperature fields

Given that the ISAS15 dataset finished at the end of 2015, the comparison with this particular dataset has been only conducted for a specific month within this timeframe. The differences in November 2015 are shown in Figures 8 and 9. Notably, there are localized disparities. The disparities take place mostly in the Arctic Ocean, in the north Atlantic coasts and Indian coast for salinity; and in the Spitzbergen area, the Baltic Sea and close by coastlines in Equatorial regions for the temperature field. These disparities can be attributed to various factors. Firstly, the dataset quality exhibits improvements when the new configuration was implemented, particularly due to the increased availability of 2015 delayed mode data in the 2020 run compared to the January 2016 run performed with the old configuration. Secondly, configuration adjustments, described in the PUM (Product User Manual), also play a role in these differences. Lastly, variations in data inputs, such as ARGO data mode change, contribute to these distinctions, which are predominantly observed in northern areas and coastal zones. Although in Figure 8 and 9 we can find multiple differences, the most significant is the ARGO data mode change mentioned before.

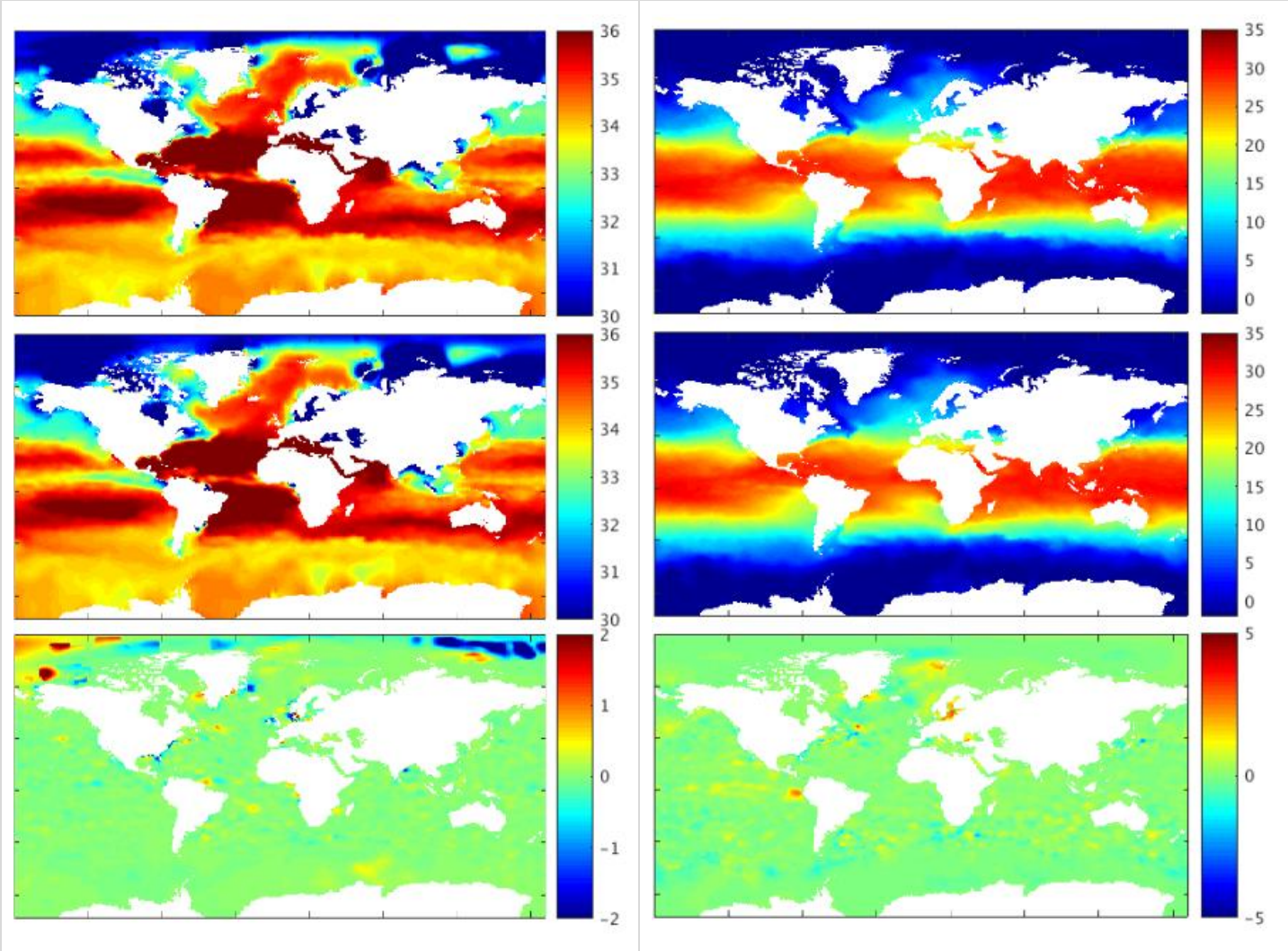

***Figure 8**: Differences in surface analysis for salinity (left panel) and temperature (right panel) fields in November 2015 between ISAS 7.0 configuration (top panels) and old ISAS 6.2 configuration (middle panels). The anomaly (measured as the difference between the new configuration minus the old configuration) is shown on bottom panels for salinity (left) and temperature (right). Units: Salinity, PSU; temperature, degrees Celsius.*

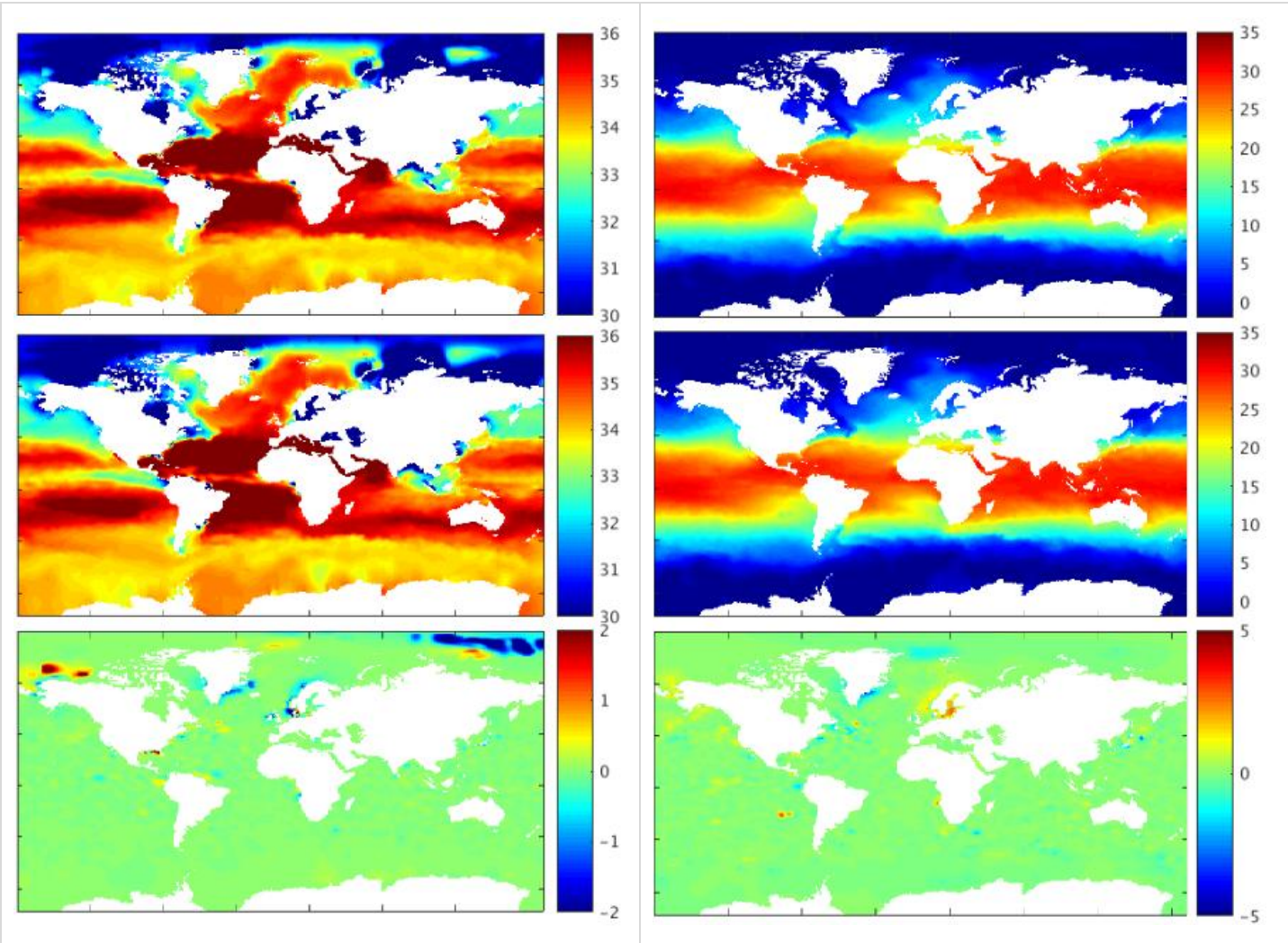

Figure 9: *Differences in surface analysed fields of salinity (left panel) and temperature (right panel) fields in November 2015 between ISAS 7.0 configuration (top panels) and ISAS15 Argo only configuration (middle panels). The anomaly (measured as the difference between the new configuration minus the old configuration) is shown on bottom panels for salinity (left) and temperature (right). Units: Salinity, PSU; temperature, degrees Celsius.*

IV.2 Global Ocean Heat Content anomaly comparison

The validation process is completed by the comparison of the global ocean heat content anomaly (GOHCA) with the estimation based on other products that are detailed here below. In this calculation, we use the INSITU_GLO_PHY_TS_OA_MY_013_052 products for the global ocean heat content anomaly (GOHCA) calculation during the 2005-2022 period and the INSITU_GLO_PHY_TS_OA_NRT_013_002 product for the calculation in 2023. Figure 10 shows that the GOHCA estimation is very close to the estimation based on SCRIPPS[3] and GCOS[10] products. It also shows a better accuracy than the ISAS 20 based estimation. The update of the validation process from 2023 to 2024 data product version has proven to give a better result for the GOHCA estimation. The global trend follows the Meyssignac et al. (2019) GOHCA trend.

[10] https://gcos.wmo.int/en/essential-climate-variables

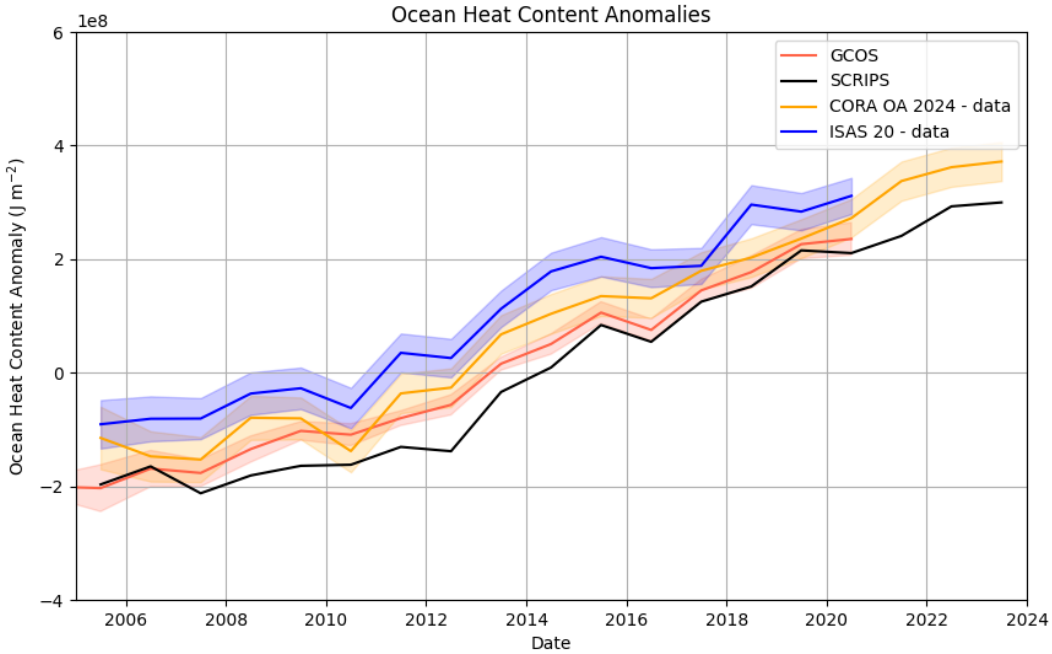

Figure 10: *Orange: Evolution of the annual mean anomaly of the global ocean heat content (GOHCA) for the objective analysis calculated with INSITU_GLO_PHY_TS_OA_MY_013_052 (2005-2022) and INSITU_GLO_PHY_TS_OA_NRT_013_002 (2023) products. Blue: GOHC estimation based on ISAS 20 product. Red: Estimation based on Global Climate Observing System (GCOS[11]). Black: GOHCA calculated with SCRIPPS products.*

[11] https://gcos.wmo.int/en/essential-climate-variables

V SYSTEM NOTICIABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to describe and track changes that may occur in the operational system. It includes changes due to outages (data gaps), upstream data (e.g., addition or loss of a sensor, inclusion/removal of in-situ platforms), reprocessing, corrections, assimilated data, boundary conditions, among other changes that may impact product quality (Table 2).

Table 2: *Changes that may impact product quality (Table 2).*

Date	**Change**	**Effects on the product**
November 2023 release	Update of the objective analysis parameters and reference fields	Better accuracy of the ocean heat content estimation on the low data coverage zones (see section IV)
November 2024 release	Vertical extension from 2000 m depth to 5500 m depth	More vertical levels (n=36) on all data and fields files from 2100 to 5500 m depth

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version (Table 3).

Table 3: *Quality changes since previous version.*

Date	Change	Effects on the product
November 2023 release	Update of the objective analysis parameters and reference fields	Better accuracy of the ocean heat content estimation in low data coverage zones (see section IV)
November 2024	No change in the data quality	No changes

VII REFERENCES

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